

WEEK 3 - VULNERABILITY ASSESSMENT

Objective:

To identify security weaknesses in the authorized lab based on Nmap scanning and network analysis using Wireshark

Findings:

1. SMB Exposure - TCP/445

Evidence:

TCP/445 was found open and identified as Microsoft-DS/SMB.

Risk:

High

Reason:

SMB provides Windows file and network-sharing functionality.

Unnecessary exposure can increase the risk of unauthorized access and information disclosure.

Recommendation:

Restrict TCP/445 using Windows Firewall and disable unnecessary SMB/file-sharing services.

2. NetBIOS Exposure - TCP/139

Evidence:

TCP/139 was found open and identified as NetBIOS-SSN.

Risk:

Medium

Reason:

NetBIOS is a legacy Windows networking service. If it is not required, exposing it increases the network attack surface.

Recommendation:

Disable NetBIOS when not required and restrict TCP/139 through Windows Firewall.

3. RPC Exposure - TCP/135**Evidence:**

TCP/135 was found open and identified as Microsoft Windows RPC.

Risk:

Medium

Reason:

RPC is used by Windows components and applications, but it does not necessarily need to be accessible from every network host.

Recommendation:

Restrict TCP/135 to authorized systems and maintain current Windows security updates.

4. Multiple Windows Services Exposed**Evidence:**

TCP/135, TCP/139 and TCP/445 were simultaneously accessible.

Risk:

High

Reason:

Multiple exposed Windows services increase the overall network attack surface.

Recommendation:

Apply firewall rules and allow only services required by authorized systems.

5. Additional High-Numbered Ports**Evidence:**

Wireshark recorded SYN/SYN-ACK communication involving ports 61728, 44505 and 52325.

Risk:

Medium

Reason:

The responses indicate additional reachable ports, although the exact applications using these ports were not identified.

Recommendation:

Identify the services using these ports and restrict unnecessary network access.

table

S.no	Finding	Evidence	Risk	Recommendation
1	SMB exposure	TCP/445 open	High	Restrict 445; disable unnecessary SMB.
2	NetBIOS exposure	TCP/139 open	Medium	Disable/restrict NetBIOS.
3	RPC exposure	TCP/135 open	Medium	Restrict RPC to authorized systems.
4	Multiple services exposed	135, 139, 445 open	High	Apply firewall restrictions.
5	High-numbered ports	61728, 44505, 52325 responded	Medium	Identify and restrict unnecessary services.

Conclusion:

The main security concerns identified were exposed Windows networking services and additional reachable ports. The assessment does not by itself prove that these services contain exploitable vulnerabilities. Firewall restrictions and service minimization are recommended to reduce network exposure.